Customer Segmentation with Actionable Business Insights

Customer Segment Report & Model Comparison

Executive Summary

This report segments the company's customer base using K-Means clustering, quantifies each segment's value with an RFM-based profile, and layers two supervised models on top: a Ridge regression that estimates a customer's total spending, and a Logistic Regression classifier that flags customers likely to respond to the next marketing campaign. All figures below come from the accompanying analysis notebook run against the full 2,240-customer dataset.

Headline findings:

- One segment (Cluster 1, 33% of customers) generates 68% of total revenue — the clear priority for loyalty and retention investment.
- A large low-spending segment (Cluster 0, 46% of customers) contributes only 7.9% of revenue and is the best fit for low-cost, automated campaigns.
- The discount-driven segment (Cluster 2, 21% of customers) shows the highest deal usage and responds well to targeted, time-boxed promotions.
- The tuned classification model reaches ROC-AUC 0.863, giving marketing a reliable way to rank customers before the next campaign rather than targeting everyone equally.
- The tuned Ridge regression explains about 80% of the variance in customer spending (R^2 = 0.803), useful for budgeting campaign spend per customer.

1. Business Problem and Objective

The company currently runs marketing campaigns without distinguishing between customer types, which risks over-discounting already-loyal customers and under-investing in high-potential ones. The objective of this project is to group customers by purchasing behaviour, describe what makes each group different, and convert those differences into specific marketing and retention actions.

1.1 Decisions This Analysis Supports

- Which customers should receive loyalty rewards versus discount-driven promotions
- Which customers are at risk of churning and need re-engagement
- Which customers are worth targeting in the next campaign (classification model)
- What a customer's expected spending is, to size campaign budgets (regression model)

2. Dataset Overview

The source data covers 2,240 customers with demographic fields (birth year, education, marital status, income, household composition), behavioural fields (recency, purchase channels, website visits, discount usage), spending by product category, and campaign response history across six prior campaigns.

2.1 Data Preparation

- 24 missing Income values were imputed with the column median.

- Unrealistic outliers were removed: ages above 90 and incomes above 200,000 (three customer records).
- Derived features were engineered: Age, Total Spending (sum of six product categories), Total Purchases (web + catalog + store), Average Order Value, Customer Tenure (days since enrollment), Household Children, and Total Campaigns Accepted.
- Categorical fields (Education, Marital Status) were one-hot encoded for the regression and classification models.
- All clustering features were standardized (StandardScaler) so that Income did not dominate the distance calculation.

3. Methodology

3.1 Segmentation (K-Means Clustering)

Clustering used seven standardized features: Recency, Total Purchases, Total Spending, Income, Average Order Value, Website Visits per Month, and Number of Deal Purchases. The number of clusters was chosen using both the elbow method (inertia vs. k) and the silhouette score across k = 2 to 10, converging on k = 4. The final model reached a silhouette score of 0.325.

3.2 Regression (Predicting Customer Spending)

Linear Regression and Ridge Regression were trained to predict Total Spending from income, age, household composition, recency, website visits, deal usage, purchase frequency, tenure, and encoded education/marital status. Ridge's alpha was tuned with 5-fold GridSearchCV over {0.01, 0.1, 1, 5, 10, 50, 100}.

3.3 Classification (Predicting Purchase Likelihood)

Logistic Regression was trained to predict Response (whether the customer accepted the most recent campaign) from recency, purchase and spending history, website visits, deal usage, income, age, household composition, complaint history, prior campaign responses, and cluster membership. Hyperparameters (C, penalty, solver) were tuned with 5-fold GridSearchCV optimizing ROC-AUC.

4. Customer Segments

The final K-Means model (k = 4) produced the following segments:

Cluster	Segment Name	Size	Revenue Share	Avg. Spending	Avg. Income	Avg. Deal Usage
0	Low-Engagement Customers	1,035 (46%)	7.9%	$103	$34,716	1.9
1	High-Value Loyal Customers	729 (33%)	68.0%	$1,265	$74,637	1.3
2	Discount-Driven Customers	471 (21%)	23.9%	$689	$54,721	4.9
3	Unclassified Outlier	1 (0.04%)	0.1%	$1,679	$51,382	0.0

Note on Cluster 3: this cluster contains a single customer with an unusually high average order value from a single purchase. It is a statistical outlier rather than a meaningful business segment and should be reviewed manually rather than targeted with a segment-level campaign.

4.1 Segment Descriptions and Recommended Actions

Segment: High-Value Loyal Customers (Cluster 1)

Highest income, highest spending, and the largest share of revenue (68%) despite being only a third of the customer base.

- Offer loyalty rewards and early access to new products.
- Introduce premium membership benefits.
- Avoid heavy discounting — these customers already show strong purchase intent.

Segment: Discount-Driven Customers (Cluster 2)

Moderate spending with by far the highest deal usage (4.9 vs. 1.3–1.9 for other segments), indicating price sensitivity.

- Send targeted, time-limited promotional offers.
- Recommend bundled products to increase order value without blanket discounts.
- Avoid discounting outside campaign windows to protect margin.

Segment: Low-Engagement Customers (Cluster 0)

The largest segment by customer count but the lowest spending and lowest income, contributing under 8% of revenue.

- Use low-cost, automated email campaigns rather than expensive advertising.
- Promote entry-level products.
- Periodically reassess whether this group should remain a marketing priority.

5. Model Performance and Comparison

Model	Objective	Baseline Performance	Tuned Performance	Selected
K-Means	Customer segmentation	Evaluated across k = 2–10	Silhouette = 0.325 (k = 4)	Yes
Linear Regression	Predict customer spending	RMSE 272.2, R^2 0.803	Not applicable (no hyperparameters)	No
Ridge Regression	Predict customer spending	RMSE 272.2, R^2 0.803	RMSE 272.2, R^2 0.803 (α tuned)	Yes
Logistic Regression	Predict purchase likelihood	ROC-AUC 0.864	ROC-AUC 0.863	Yes

5.1 Regression — Predicting Customer Spending

Ridge Regression achieves an R^2 of 0.803 and a mean absolute error of about $200, meaning the model explains roughly 80% of the variance in customer spending using only demographic and behavioural features (no spending data itself). Tuning alpha produced essentially the same accuracy as the baseline, indicating the baseline regularization strength was already close to optimal for this feature set.

5.2 Classification — Predicting Purchase Likelihood

Only 14.9% of customers accepted the most recent campaign, so the model was evaluated with precision, recall, F1, and ROC-AUC rather than accuracy alone. The tuned model (C = 0.1, L2 penalty, liblinear solver) reaches ROC-AUC 0.863, precision 0.66 and recall 0.28 for likely responders. In practice this means the model is conservative: when it flags a customer as likely to respond, it is right about two-thirds of the time, but it misses a meaningful share of true responders. This profile fits a budget-constrained campaign — it identifies a smaller, higher-confidence pool of high-potential customers rather than casting a wide net.

6. Business Insights and Recommendations

1. Highest revenue segment: High-Value Loyal Customers (Cluster 1) — protect this relationship with loyalty perks and early access rather than discounts.
2. Most price-sensitive segment: Discount-Driven Customers (Cluster 2) — the highest deal usage in the dataset; use time-boxed promotions and bundles instead of standing discounts.
3. Lowest priority for paid marketing spend: Low-Engagement Customers (Cluster 0) — nearly half of customers but under 8% of revenue; use low-cost channels only.
4. Campaign targeting: rank customers by the tuned classification model's predicted probability and set a probability threshold based on campaign cost versus expected revenue, rather than emailing the full customer base.
5. Budgeting: use the tuned Ridge regression's spending estimate to size expected order value per customer when planning campaign economics.

6. Data quality follow-up: investigate the single-customer outlier cluster (Cluster 3) and the wider Income/Age outliers removed during cleaning — they may indicate data-entry errors worth correcting at the source.

7. Conclusion

Segmenting customers into four groups — with one flagged as a data outlier rather than a true segment — reveals a highly concentrated revenue base: one-third of customers generate two-thirds of revenue, while nearly half of customers contribute less than 8%. Pairing this segmentation with a spending-prediction model ($R^2 = 0.803$) and a campaign-response classifier (ROC-AUC = 0.863) gives the marketing team both a map of who its customers are and two working tools to act on that map: budgeting by predicted spend, and targeting by predicted campaign response.

Recommended next step: pilot the segment-specific actions in Section 6 on a small, held-out group of customers per segment, and measure incremental revenue and response rate before rolling out company-wide.